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An Intelligent Heart Disease Prediction Model using Adaptive Multiscale Neural Network with MSE and Categorical Cross Entropy Loss Functions

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Abstract

Cardiovascular diseases remain the primary cause of global mortality, necessitating advanced diagnostic tools for early intervention. Heart rate prediction using Electrocardiogram (ECG) signals serves as a vital biomarker for identifying life-threatening cardiac irregularities. By leveraging deep learning and bio-inspired optimization, researchers can achieve high-precision monitoring to improve patient outcomes. Thus, this work developed an enhanced deep learning model with a novel loss function for heart disease prediction, incorporating optimization mechanism. Initially, ECG signals are collected from the benchmark datasets and given to the feature extraction process. Here, three types of features are extracted, where Spatio-Temporal Attention-based Sparse Autoencoder (STA-SAE) extracts the first set of features (F1), spectral features are indicated by (F1), and wave features are (F3). Then, these three significant sets of features are given to the Adaptive Multiscale Elman Recurrent Neural Network with Novel Loss (AMS-ERNN-NL)-based heart disease prediction model. Here, the multi-scale element aids in processing the features at three different scales (scale-1 for F1, scale-2 for F2, and scale-3 for F3). Then, the multi-scaled features are concatenated and given to the Elman Recurrent Neural Network (ERNN), where it has the ability to remember the past state, thus allowing the AMS-ERNN-NL model to capture the dynamic and evolving conditions of the patient's health. Moreover, a novel loss functions, named Mean Squared Error (MSE) and categorical cross entropy and adaptive strategy is infused with it to further enhance the effectiveness of the developed model. Here, the adaptive strategy allows for parameter optimization concept, where it is performed using the Fitness Refined Randomized Starfish Optimization Algorithm (FRR-SFOA) to tune parameters, leading to reduced overfitting. Finally, by using the FRR-SFOA, the AMS-ERNN-NL offers the predicted outcome.

Keywords Heart Disease Prediction · Electrocardiogram · Fitness Refined Randomized Starfish Optimization Algorithm · Adaptive Multiscale Elman Recurrent Neural Network with Novel Loss



1 Introduction

The heart has become a significant part of the body, which aids in delivering nutrients and oxygen to the tissues present in the body and also helps to remove the waste particles [1]. The total number of times the heart is beaten is the "heart rate", where the unit is beats per minute (bpm) [2]. By manually checking the pulse using different kinds of consumer and medical devices, the heart rate will be measured [3]. For adults, the normal heart rate is 60-100 bpm, according to the American Heart Association. Yet, this range can differ depending on the parameters, such as level of fitness, health and age. The heartbeat can be controlled by employing the electrical conductivity mechanism, which aids in directing the heart chamber to relax and contract within a coordinated rhythm. This rhythm has ventricles, sinoatrial and atrioventricular modules, where they support passing the heart's signaling to all the parts of the body [4]. The heart rate detection allows for the determination of different heart diseases, including Arrhythmias and Coronary Artery Disease (CAD) [5].

Different tools have been designed to detect the heartbeat rate [6], where some tools, such as chest straps, oximeters and fitness trackers. Moreover, the clinical experts have performed auscultations to detect the heartbeat, while the clinical actions of the heart are detected through ECGs [7]. It offers an accurate heart rate measure, as well as the rhythm [8]. From the ECG signals, heart rate is calculated by tracing the interval between consecutive R-waves, which has become a tall spike, indicating the ventricular contraction [9]. The existing models are used to detect whether the heart rhythm is irregular or regular [10]. They pave the way to identify even the complicated ECG patterns related to the heart rate based on the prehistoric manual characteristics which are difficult to comprehend [11]. Further, such systems are sometimes prone to different types of artifacts and noises and have the complexity in detecting the non-linear patterns that are used to characterize the different cardiac abnormalities [12]. The traditional mechanism requires more time to provide accurate computation, which hinders the system's efficacy.

Recently, deep learning models have made significant contributions to detecting the heart rate using ECG patterns, thus allowing the models to offer automotive feature retrieval and also support tackling the generalization issues [13]. They have the ability to offer continuous processing of data that is present in the ECG signals by generating the effective heart rate detection results [14]. By learning the more robust and resilient noisy and artifact data, the deep learning models can provide enhanced results [15]. These networks can handle various signal sets that are correlated with the ECG and provide successful interpretations [16]. In addition to all these significances, they have the capacity to process the data by considering long-term and local characteristics, resulting in effective interpretation of spatial and temporal data [17]. Therefore, this research work develops a novel deep learning approach towards detecting the heart rate using ECG signals by overcoming the major diminishing threats found in the pre-existing approaches, and the section that follows provides the model's contribution.

- To develop an innovative deep learning based heart disease prediction system by leveraging the power of an adaptive optimization approach with a feature extraction process, thus enhancing the diagnostic reliability. It aids in processing the ECG signals via the spatio-temporal, spectral and wave-based feature representation, thus allowing the proposed model to provide a deeper understanding of the complicated features. By the combination of optimization algorithm with the deep learning model, the proposed system attains greater robustness and adaptability towards the varying ECG databases, thus establishing a strong foundation to offer precise and effective prediction, providing its enhanced sensitivity.
- To suggest the FRR-SFOA for optimizing the parameters of the developed deep learning-based prediction model. It leverages the principle of traditional Starfish Optimization Algorithm (SFOA) and incorporates a fitness-driven refinement strategy with it to design the enhanced version, named FRR-SFOA. It aids in providing adaptive exploration and exploitation balance by employing the randomized mechanism, thus preventing the model from early convergence. IT tunes the hidden neurons, epochs and learning rate to improve the Matthews Correlation Coefficient (MCC) and precision. The refined strategy also leads to R-SFOA to enhance the

stability, reduce the overfitting issue, and provide consistent diagnostic accuracy, thus making it an intelligent optimizer for prediction tasks.

- To design the AMS-ERNN-NL for predicting heart disease using ECG signals. The multiscale element incorporated in the developed AMS-ERNN-NL aids in handling the wave, spectral and spatio-temporal patterns at three distinct scales, thus ensuring that the developed model learns the comprehensive feature representation. To capture the long-term cardiac patterns, the ERNN is used, while the novel loss function, named MSE and categorical cross entropy helps to increase the sensitivity and stability of the model towards both the subtle and major abnormalities. It combines the advanced optimization, FRR-SFOA, which is used to attain enhanced robustness, generalization and accuracy by optimizing the critical ERNN parameters, proving the strong clinical predictive potential.

This section shows the work's organization. Need for the heart disease prediction model and its novel descriptive illustration using the heart disease ECG dataset has been offered in Section III; Multi-step feature extraction process based on spatio-temporal, spectral and wave features is shown in Section VI; Development of novel adaptive multiscale Elman network for heart disease prediction with novel loss and optimized parameters is detailed in Section V; results and discussion are given in Section VI, and conclusion is given in Section VII.

2 Literature Survey

2.1 Related Works

In 2023, Tahmid et al. [18] have developed a Multi-Dimensional Deep Neural Network (MD)-CardioNet to capture the volumetric, temporal and spatial features given in the ECG through the application of multi-dimensional convolutions. The practical knowledge distillation model was also here to reduce the requirement of more trainable variables by maintaining accurate results. The results showcased the enhanced robustness of the MD-CardioNet model.

In 2024, Ayano et al. [19] have presented an innovative Gradient-weighted Class Activation Mapping++(Grad-CAM++) to estimate the heartbeat. It incorporated a multi-scale hybridized structure, integrating 12 numbers of blocks. The model provided the enhanced outcome and was trained deeply to visualize the powerful ECG segments. Based on the results, the developed model makes it a valuable choice for real-time implementation.

In 2024, Naeem et al. [20] have presented an Artificial Neural Network (ANN) to detect heart disease. By extracting the sensor data, the ANN exhibited superior results. The measures, like specificity, accuracy and f1-score, was used for the validation, and offered 85% of accuracy. This work thus proved the power of a feature extraction system in detecting diseased individuals.

In 2023, Lee et al. [21] have proposed a single-lead ECG measurement model, which could detect the heart status accurately. It incorporated four different models and 13 features. It provided the best performance by attaining a time of 0.86 s.

In 2024, Raoof et al. [22] have developed a Visual Geometry Group (VGG16) model to detect the class of cardiac disease automatically. More analyses have been done to evaluate the supremacy of the system, while the results have proven that it enhanced the prognostics and detection ability.

In 2025, Hasan et al. [23] have used eXtreme Gradient Boosting (XGB), Gaussian Naive Bayes (GNB), Random Forest (RF) and Decision Tree (DT) to estimate heart disease. To retrieve the relevant features, a deep learning model was applied effectively. Weights were tuned to improve the working performance. It achieved 99.29% of accuracy, outscoring the baseline systems.

In 2024, Utsha et al. [24] have introduced a Long Short-Term Memory (LSTM) for monitoring the heart rate. It could visualize the EEG patterns effectively by providing optimal tracking. A custom wearable device was

infused to transfer the collected data in real-time. The recall, precision and accuracy of this model were 99.86%, 99.95% and 98.74%, proving its adaptability.

In 2024, Prusty et al. [25] have presented a Convolutional Neural Network-based Scale-Invariant Feature Transform CNN (SIFT-CNN) for the classification of heart disease. It classified the disease into three different types, and they were arrhythmia, normal sinus rhythm and congestive heart failure. It attained 99.78% of the F1-score. The SIFT–CNN was exceptionally sound and better at classifying heart diseases.

2.2 Problem Statement

Estimation of heart rate enables the interpretation of various heart-related health ailments. However, the conventional models associated with the estimation of heart rate find major trouble in identifying the complex features present in the heart signals by missing out the fine-grained underlying details, thereby resulting towards false negatives. Further, these approaches are difficult to adapt to a wide range of modalities due to the presence of heterogeneous, fragmented data availability and raise a major concern towards their clinical interpretations. By considering all these aspects, this research work has framed a novel deep learning approach towards the detection of heart rate using ECG signals. Table 1 shows the major features and challenges present in the traditional approaches, and the following section discusses the major research gaps present in the pre-existing models.

- The data present in the ECG signals are heterogeneous and incompatible in nature. For processing these kinds of data resources, the traditional models are not as effective as they primarily utilize a fixed set of features for enabling interpretations, and they lead to poor generalized outcomes. Further, these techniques may fail to consider the subtle, non-linear patterns present in the ECG signals. Hence, there is s need for the enforcement of a novel deep learning approach that enables the effective interpretation of fine-grained patterns present in the heart signals and provides precise heart rate estimations.
- Another challenge is that the noise in ECG signals is extremely sensitive, which can influence the precision of extracted features for identifying cardiovascular disease from ECG. To overcome such challenges, a feature extraction process was utilized. The effective feature extraction process separates noise in the ECG signals and maintains crucial structural characteristics.
- Conventional methods may ignore the minor defective regions they resulting towards false negative predictions. Further, these methods are subjected towards higher cognitive loads, as they possess noisy regions and involve a huge time for enabling precise computations. In order to overcome these defects, research is mandatory towards incorporating a sparse autoencoder approach that helps in overcoming the defects generated due to biased estimations by handling multimodal data features present in the heart patterns and enabling the effective interpretation of imbalanced data.
- The existing models developed for estimating heart rate associated with ECG signals fail to generate the optimal solutions, as the detection parameters may attain a sub-optimal value when operated for a prolonged period. As a result, these models may fail to explore the wider available regions present in the global search spaces and fail to converge on the local optimal regions. Hence, this research should focus on the utilization of a novel optimization algorithm that should enhance the working performance of the heart rate estimation model towards attaining accurate estimations by rapid convergence of local optimal regions.

Table 1 Features and Challenges Present in the Traditional Heart Rate Prediction Approaches

Author [citation]	Methodology	Features	Challenges
Tahmid et al. [18]	MD-CardioNet	• This approach poses comparatively lower false alarm rates than other methods • This method categorises the ECG signals using a binary classification approach	• This technique may fail to deal with the heterogeneous data • This method may tend to overfit in the case of limited datasets
Ayano et al. [19]	Grad-CAM + + and SHAP	• This technique has extracted the temporal and spatial relationship associated with the ECG signals • The attention module present in this model has provided weightage to the most relevant features	• This model fails to handle the class-imbalanced datasets • The workability of this method may be affected due to scalability issues
Naeem et al. [20]	ANN	• This method expands the training data by involving definite data augmentation approaches • The ECG signals obtained using this method is converted into spectrograms for extracting the prominent features	• The presence of noisy artifacts could influence the effectiveness of this model • This method struggles with vanishing gradient issues
Lee et al. [21]	Single-lead ECG Measurement System	• The ECG waveforms generated using this approach are reconstructed to interpret the complex intervals • The incorporation of a loss function using this technique enhances the quality of the extracted ECG signals	• The real-time deployment of this method may require huge computational costs • This method may be affected due to signal overlapping issues, due to the presence of multiple artifacts
Raoof et al. [22]	VGG16	• This approach significantly diminishes the vanishing gradient issues through the presence of skip connections • The size of the training data is expanded to improve the scalability of this model	• This method is influenced due to imbalanced data classes • The long-term temporal data are not fully captured using this approach
Hasan et al. [23]	GNB, XGB, DT, RF	• This method reconstructs the ECG signals into latent spaces of lower dimensions • A single vector representation formed by this approach captures the ECG signals effectively	• The inference time possessed by this model is comparatively larger • The presence of artifacts could interrupt the standardization ability of this approach
Utsha et al. [24]	LSTM	• The noise and artifacts present in the ECG signals obtained using this model are simplified by the presence of bandpass filters • This stem generates a beat-by-beat plot representation of the ECG signals for enabling effective estimations	• This method possesses a higher number of false positive predictions • This technique contains minimal contextual information, which further affects the transparency of this approach
Prusty et al. [25]	SIFT–CNN	• The signals extracted using this technique are normalized for obtaining standardized heart rate estimations • Low contrast key points are filtered out effectively using this approach	• The ECG signals converted into images could lead towards the loss of fine-grained details • This approach may lead to overfitted results when dealing with limited datasets

3 Need for the Heart Disease Prediction Model and its Novel Descriptive Illustration Using the Heart Disease ECG Dataset

3.1 Need for the Heart Disease Prediction Model Using Deep Learning

Since the burden of heart disease has become increasing, it requires designing an accurate, fast and reliable system to provide enhanced prediction results to help doctors in offering early risk evaluation and diagnosis. The performance of the existing systems is based on the interpretation of the clinicians, handcrafted features and statistical risk calculators; thus, they may not be effective in capturing the non-linear and subtle variations that exist in the physiological signals, like EEG. However, these issues can be tackled by applying effective deep learning models. They have the ability to learning the more complicated patterns automatically, directly from the less processed data input. This enables the models to detect the hidden cardiac trends that are not easily processed by the traditional systems. Furthermore, the existing models, which are based on deep learning, are effective in handling all kinds of data, even in the presence of noisy and high-dimensional elements, towards different populations. By considering all these advanced features, this work developed an innovative model by leveraging the deep learning model, which aids in designing an automotive and robust network to support real-time decision-making, thus enhancing the outcome of the patients.

3.2 Descriptive Illustration of the Designed Heart Disease Prediction Model

This work suggested an improved deep learning model for predicting heart disease through the incorporation of a novel loss function and optimized mechanism. It has the capability to analyze the ECG signals more effectively by capturing simple and more complicated cardiac representations. The proposed model can also handle even the noisy data with greater stability, along with the real-world data. By collecting the required ECG signals from the standard data sources, the prediction process begins, which ensures that the collected input represents a wide variety of normal and abnormal cardiac patterns. Then, such collected signals are provided to the multi-stage feature extraction phase, where three sets of features, including spatio-temporal, wave, and spectral features, are extracted. This process aids in capturing the full cardiac spectrum information. In this process, STA-SAE is used to extract the spatio-temporal features, indicated by F1. Then, the second and third sets of features are spectral and wave features, indicated by F2 and F3, respectively. Once three significant sets of features are extracted, they are further given to the AMS-ERNN-NL model for the heart disease prediction process. Here, the multiscale element initially receives the obtained feature in its three varying scales, i.e., scale-1 for F1, scale-2 for F2, and scale-3 for F3, thus allowing the proposed model to simultaneously learn the global, intermediate, as well as local waveform features. Then, the resultant features are combined and given to the ERNN model to provide the final prediction result. To retain the contextual data, it employs the internal context layer, thus allowing the system to capture the continuous cardiac changes and growing abnormalities. Moreover, this model also integrates the novel loss function, like MSE and categorical cross entropy, which aids in balancing the network sensitivity towards the huge deviation and subtle variation, thus enhancing the generalization. The performance of this model is enhanced by leveraging the optimized parameters, using the FRR-SFOA, which optimizes the hidden neuron, learning rate, and epoch, resulting in reduced overfitting. Through the combined strengths of multiscale learning, novel loss formulation, and FRR-SFOA-guided optimization, the AMS-ERNN-NL model delivers highly reliable and precise heart disease predictions across diverse ECG datasets. A diagrammatic illustration of the designed heart disease prediction model is given in Fig. 1.

3.3 Heart Disease ECG Dataset

The ECG signals necessary to perform the heart disease classification are taken using the dataset link of https://www.kaggle.com/datasets/shayanfazeli/heartbeat. Access date: 2025–12-15, where the dataset is named “ECG

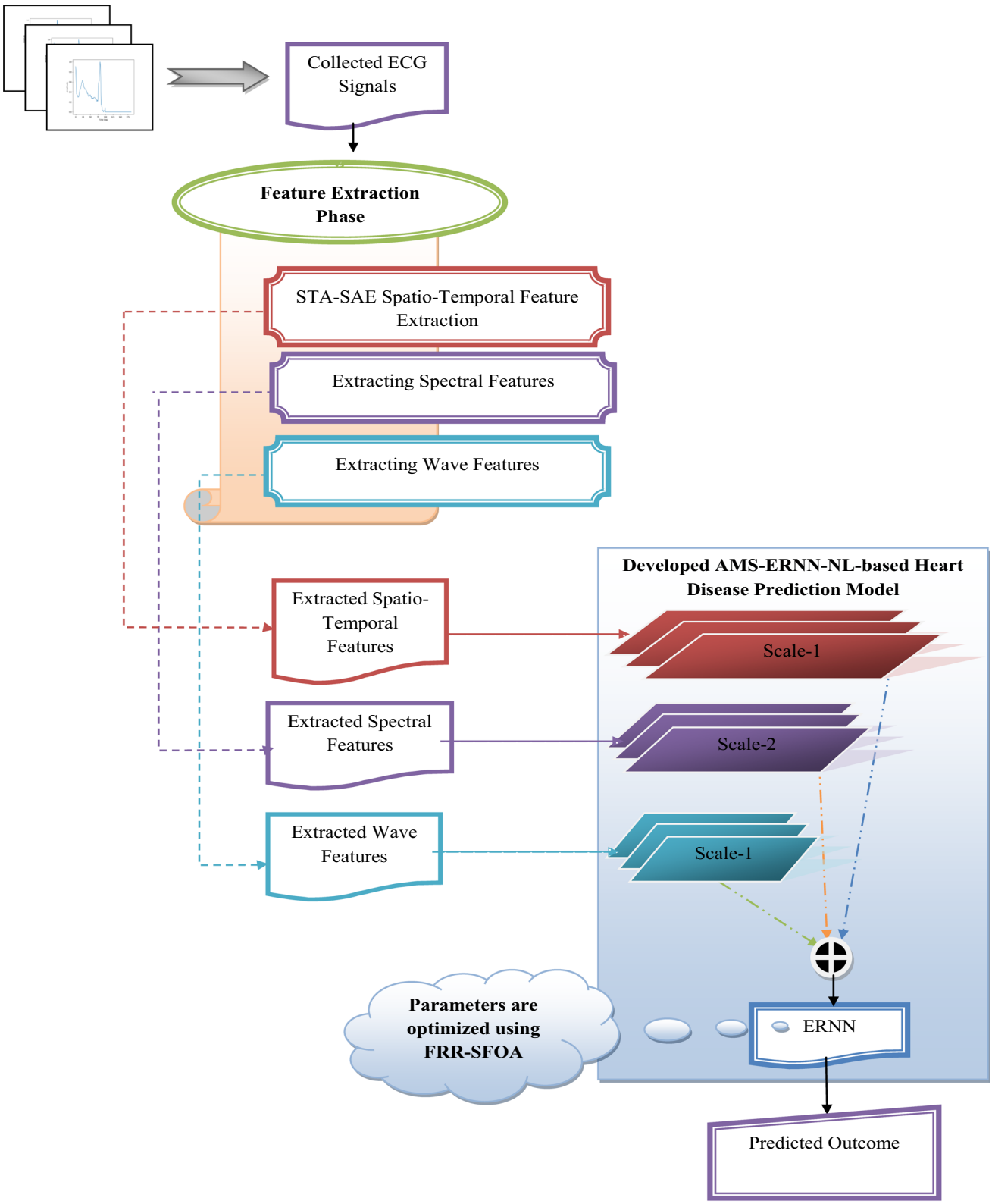


Fig. 1 Diagrammatic Illustration of the Designed Heart Disease Prediction Model

Heartbeat Categorization Dataset". This dataset has 5 classes, and 125 Hz Sampling Frequency. It includes 4 major files, which are named (i) mitbih_test.csv, (ii) mitbih_train.csv, (iii) ptbdb_abnormal.csv, and (iv) ptbdb_normal.csv, incorporated with 752 columns, where the size of this dataset is 582.79 MB. The collected ECG signals are indicated by the term Sig^{ECG}, and some of them are given in Fig. 2.

4 Multi-step Feature Extraction Process based on Spatio-temporal, Spectral and Wave Features

4.1 Feature-1: STA-SAE-based Features

The collected ECG signals Sig^{ECG} are given to the STA-SAE, which aids in extracting the feature set-1, named spatio-temporal features. The STA-SAE is developed by the integration of a sparse autoencoder with the spatial and temporal attention network. It performs by concentrating on both the temporal and spatial trends present in the given inputted data.

Initially, in the sparse autoencoder [26], the autoencoder is used to retrieve only the most suitable features given as input by leveraging the sparse element through the application of numerous hidden neurons. The activation operation for the used hidden neuron is $l_n\,(b)$, while the term b is the input and the averaging activation function is $\hat{\Re}_n$. This process is given in Eq. (1) mathematically.

$$\hat{\Re}_n = \frac{1}{q}\sum_{m=1}^{q}\left[l_n\left(b^m\right)\right] \tag{1}$$

Then, based on Eq. (2), the hidden neuron is set to 0 for further processing.

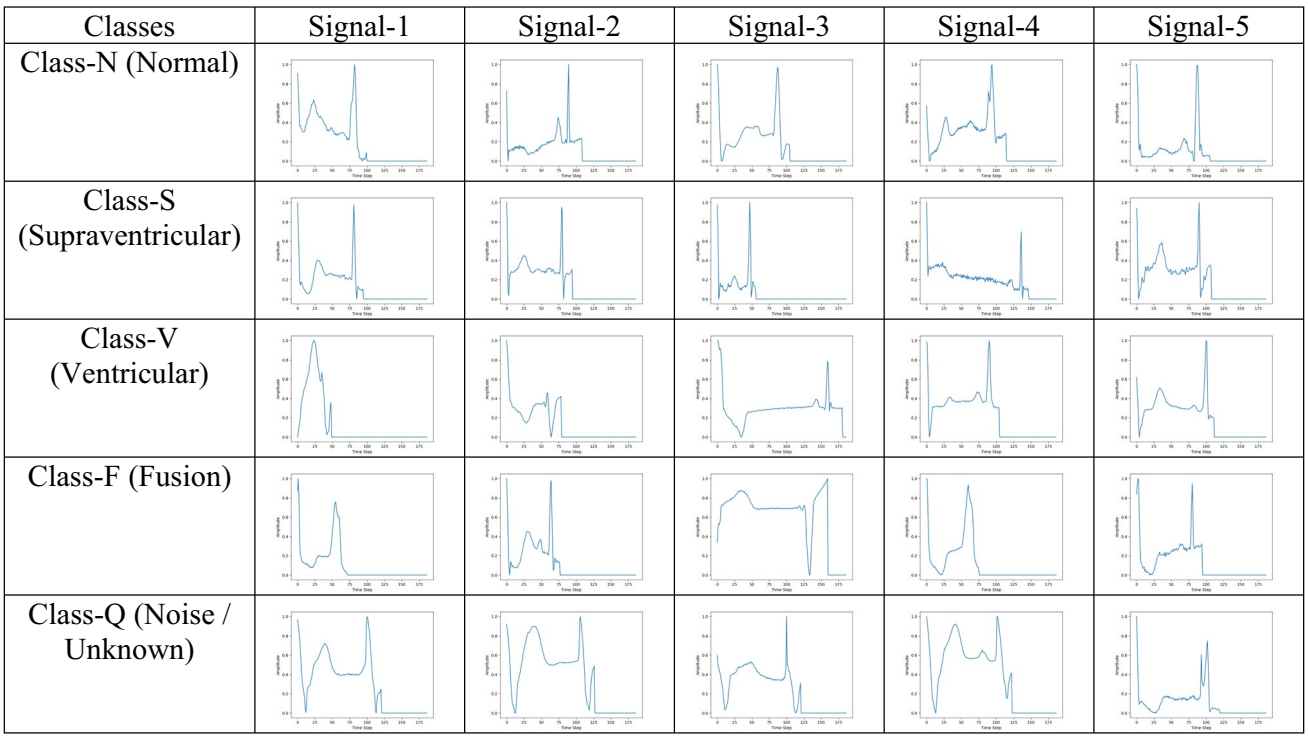


Fig. 2 Collected ECG Signals

$$\hat{\Re}_n = \Re \tag{2}$$

The term $\Re$ is the sparse parameter. By setting the limit to 0, this sparse unit aids in extracting the relevant features. Yet, the sparse autoencoder has the complexity in effectively capturing the complicated temporal and spatial data, which are present in the essential patterns. Thus, the STA [27] is added with the SAE, helping to solve the issue by the extraction of complicated features.

This two-phase mechanism aids in ensuring that the developed model learned the relevant and informative features.

To extract the spatial patterns present in between the exogenous and historical target series, the spatial element can be applied, while the temporal block is used to extract the long-term temporal dependencies. Here, the size of the window is given as X, while the input is $b_x = B_x^{(1)}, B_x^{(2)}, ..B_x^{(r)}$, the dimension of the exogenous series is r, and its time step is x. Further, with the initial input data, the linear transformation is performed, thus generating the weight vector for the spatial block, indicated as g_x, and it has been numerically validated by Eq. (3).

$$g_x = A_g^X b_x + f_g \tag{3}$$

The parameter in Eq. (3) is $A_g^X b_x$, and the weighted attention factor is f_g, and then such a factor has been normalized through the softmax function to get the enhanced factor leveraging Eq. (4).

$$\aleph_x^{(o)} = \frac{\exp\left(g_x^{(o)}\right)}{\sum_{m=1}^{r+1} \exp\left(g_x^{(m)}\right)} \tag{4}$$

Here, the term m is the size of the window, where it creates the temporal attention weightings $h^{(m)}$, thus enhancing the exogenous series' efficacy using Eq. (5).

$$h^{(m)} = A_h^X b^{(m)} + f_h \tag{5}$$

The normalized and extracted parameters are $\alpha_x^{(m)}$ and $A_h^X b^{(m)}$, where the term $\alpha_x^{(m)}$ can be determined through Eq. (6).

$$\alpha_x^{(m)} = \frac{\exp\left(g_x^{(l)}\right)}{\sum_{x=1}^{X} \exp\left(g_x^{(m)}\right)} \tag{6}$$

The current time step is $x \in [1, X]$. Thus, the ATA-SAE provided the extracted spatio-temporal features $Fe1^{STASAE}$ as the final outcome. By the integration of integrating STA-SA, the proposed model enhanced the aptitude to distinguish complicated patterns and background dependencies. STA-SAE-based Feature extraction model is pictorially viewed in Fig. 3.

4.2 Feature-2: Spectral Features

The collected CEG signals Sig^{ECG} are used in this phase for extracting the spectral features. Extracting the spectral features aids in converting the EEG signals from the time domain to the frequency domain, thus capturing the frequency and rhythm-dependent hidden features in the raw waveforms. Through applying the Short-Time Fourier Transform (STFT), the signals will be decomposition to their frequency components, thus allowing identifying the abnormal classes in terms of harmonic content, periodicity, as well as the distribution of power. Finally, the extracted spectral features $Fe2^{Spectral}$ are useful to detect the ischemic changes and the variation present in the heart rate.

 https://doi.org/10.1007/s44196-026-01327-7

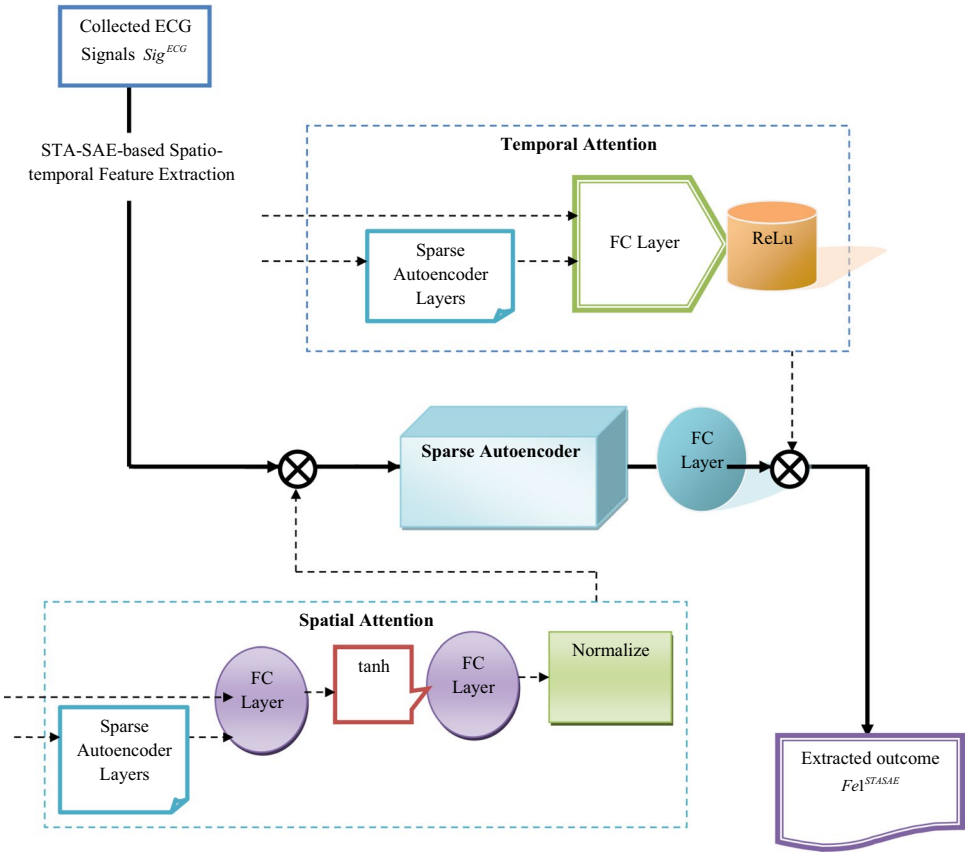


Fig. 3 Pictorial View of the STA-SAE-based Feature Extraction Model

4.3 Feature-3: Wave Features

From the collected signals Sig^{ECG}, extracting the wave features helps to capture the EEG's morphological structure through the analysis of waveform characteristics, which include:

- P-wave
- QRS complex
- T-wave

The above-mentioned elements can reflect the essential electrophysiological incidents that are present in the cardiac cycle, amplitude variations, slope, shape, and durations, where these are the robust and fundamental cardiac pathology indicators. Extracting such features also acts as a main role in detecting the ischemic changes, arrhythmias and conduction abnormalities, thus offering an enhanced and richer representation of the data that complements spatio-temporal and spectral features in building a robust and comprehensive prediction model. Finally, the extracted wave features $Fe3^{wave}$ are obtained.

5 Development of Novel Adaptive Multiscale Elman Network for Heart Disease Prediction with Novel Loss and Optimized Parameters

5.1 FRR-SFOA for Performance Enhancement

The FRR-SFOA is developed for optimizing the parameters in the AMS-ERNN-NL-based heart disease prediction model. It plays a major role and helps to strengthen the efficacy of the designed system.

Novelty: The novelty of the developed FRR-SFOA is present in its refined and adaptive randomized concept, formed by leveraging the fitness scores. It intelligently tunes the internal learning parameters that govern the temporal dynamics and multiscale processing capabilities of the network. The traditional SFOA [28] updates the position according to the arm extension and fundamental radial motion, but the proposed FRR-SFOA solution uses the random number refinement mechanism, which aids in elevating the adaptability and stability during the optimization process. In this novel enhancement, the fitness-based refinement concept is used, where the random number $Rand$ in the traditional SFOA gets amended by employing the current, worst, mean and best fitness values given in Eq. (7).

$$Rand = \frac{mF + bF}{cF + mF + wF + bF} \tag{7}$$

Here, the terms mF, wF, bF and cF are the mean, worst, best and current fitness values. This adaptive randomization helps to adjust the exploration power of the FRR-SFOA in accordance with the real-time searching landscape, thus allowing the optimizer to extend the searching radius if there is low diversity and tighten the searching process if there is a nearly placed and high-quality solution. By injecting the controlled stochastic perturbations, this randomized strategy has become a novelty, thus helping to prevent stagnation by ensuring that the FRR-SFOA does not fall into premature convergence throughout the sub-optimal area, which is a common issue encountered in conventional systems due to the highly nonlinear ECG patterns.

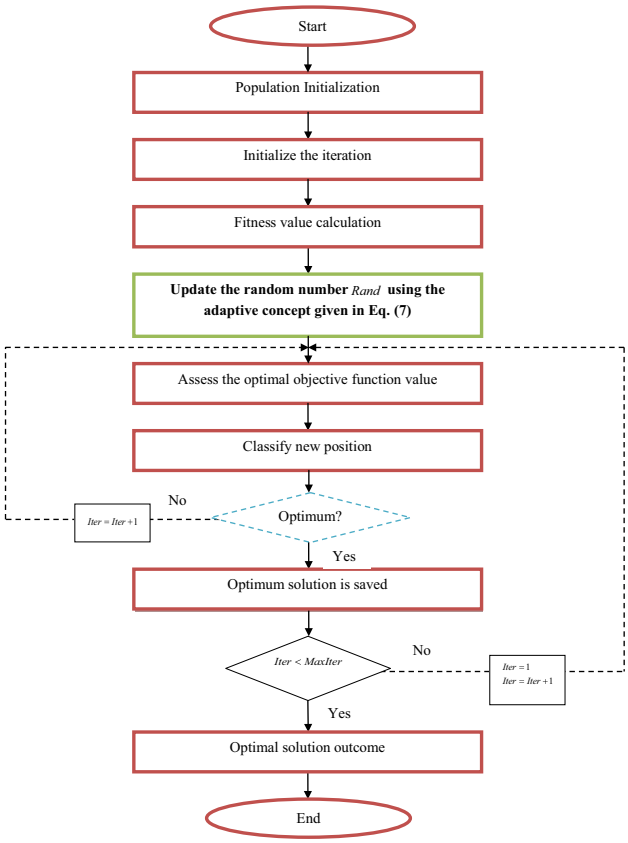
Through the refined randomized mechanism, the developed FRR-SFOA helps effectively to tune the parameters, such as hidden neuron count, learning rate, and epoch count, by assigning various updating intensities to these parameters, thus helping to identify their unequal sensitivity and impact on ERNN temporal learning. By tuning these parameters in the AMS-ERNN-NL framework, the AMS-ERNN-NL framework generates the population iteratively and evaluates the candidate configuration to attain the objective, like enhanced precision and MCC.

Using the updated random number by considering the fitness scores, the FRR-SFOA helps reduce the bias and discriminative power of the system, which may result in reduced overfitting, improved gradient stability, rapid convergence and superior ability across the meaningful ECG patterns. Moreover, the interaction present in between the loss function and the optimized parameters using the FRR-SFOA enhances the ability of the model, thus helping to capture subtle changes. Figure 4 offers the flowchart of the proposed FRR-SFOA.

5.2 Integration of Novel Loss Functions: MSE and Categorical Cross Entropy

Using the loss functions, such as MSE and categorical cross entropy helps to balance the sensitivity of both subtle and larger prediction errors throughout the learning stage. The existing models have used some loss function, which emphasize only one kind of derivation, like minor variation focusing or amplification of huge discrepancies, leading to limited generalization ability and biased gradient. But the loss function formulation used in the developed model can overcome the existing issues through the integration of features of different error metrics into a single and unified element. This design also strengthens the responses of the model inherently by maintaining the model's ability to identify the fine-grained variations that occur frequently in the complicated EEG patterns. Moreover, categorical cross entropy function has also been used in the loss function, which aids in adjusting the magnitude relative to the size of the dataset by ensuring a stable learning dynamic and preventing the overwhelmed gradients. Further, it also improves the converging behaviors of the optimization approach and its robustness towards the signal noise by promoting more effective data representation. Thus, the developed model has become more effective in attaining clinically relevant and consistent results, even when trained on heterogeneous or noisy ECG datasets.

Fig. 4 Flowchart of the Proposed FRR-SFOA



5.3 Development of Novel AMS-ERNN-NL for Heart Disease Prediction

The AMS-ERNN-NL is developed for the heart disease prediction process, which is designed by leveraging the multi-scale component, a deep learning network (ERNN) with optimized parameters using the FRR-SFOA.

Novelty: The implementation of an innovative AMS-ERNN-NL is based on a unified and highly effective model, which has the ability to capture the complicated and multifaceted trends that are present in the EEG signal patterns. Initially, the process is initiated by extracting the three significant feature sets, like spatio-temporal $Fe1^{STASAE}$, wave $Fe3^{wave}$ and spectral descriptors $Fe2^{Spectral}$ from the collected EEG signals. Using a single-scaled network to process these given features may result in losing essential details, as the set of each feature has unique diagnostic information. Thus, the multi-scale-based AMS-ERNN-NL network is developed in this work for the prediction process. The attained three sets of features are initially processed by the multi-scale element, which introduces three scales to process the unique features independently, where they are used to handle different magnitudes of features and temporal variations. Scale-1 captures the spatio-temporal patterns $Fe1^{STASAE}$, emphasizing the fine-grained local abnormalities, and scale-2 detains the spectral features $Fe2^{Spectral}$, indicating the early pathological changes, while scale-3 wave features $Fe3^{wave}$ to describe the overall cardiac rhythm.

Then, the features processed by the three scales of the multiscale element are further fused and given to the ERNN to provide final prediction results. By incorporating the context memory layer, the ERNN can retain the past activation. This allows the proposed AMS-ERNN-NL to effectively learn the long-term patterns, thus making the model suitable and optimal for detecting the cardiac conditions that unfold over multiple ECG cycles. The feedforward neural network present in the ERNN [29] forms the basis. Generally, the ERNN has three significant horizontal layers, which are indicated by the terms x,n, and v. It also has a background element p in the hidden layer. Then, the given fused input and the hidden components are placed inside the backpropagation layer, where they employ a sigmoid coefficient producing real values, ranging from [0,1] by using Eq. (8).

$$n = \frac{1}{(1 + m - sum)} \tag{8}$$

Some factors are considered in the backpropagation, which include discrimination and transfer function, learning velocity and velocity rate. Thus, the complexity may be increased, since the model can be impacted greatly by these factors. Thus, an adaptive learning mechanism has been integrated into the AMS-ERNN-NL as an innovative element. The existing optimization approaches are widely employed for the detection process by leveraging the fixed set of parameters, which may result in sub-optimal convergence. But the developed model incorporates an adaptive parameter optimization strategy, named FRR-SFOA, by leveraging the principle of pre-existed SFOA, resulting in enhanced convergence to attain the optimal solution quicker. By tuning the network parameters, like hidden neurons, epochs, and learning rate, this developed FRR-SFOA aids in ensuring that the learning trajectory remains stable even when the input ECG signals are noisy or highly variable across subjects. Equation (9) shows the objective function ObJ formulation of the developed model.

$$ObJ = \underset{\{H_Q^{ERNN}, SE_V^{ERNN}, BS_G^{ERNN}\}}{\arg\min} \left(\frac{1}{PRE + MCC}\right) \tag{9}$$

Here, the hidden neurons H_Q^{ERNN}, epochs SE_V^{ERNN}, and learning rate Bl_G^{ERNN} of ERNN are optimized in the range of $[5, 255]$,$[5, 50]$ and $[0.01, 0.99]$, respectively.

- Precision PRE is determined by Eq. (10).

$$PRE = \frac{c_x}{c_x + e_z} \tag{10}$$

- Matthews's correlation coefficient (MCC) MCC is determined via Eq. (11).

$$MCC = \frac{c_x \times d_y - e_z \times f_a}{\sqrt{(c_x + e_z)(c_x + f_a)(d_y + e_z)(d_y + f_a)}} \tag{11}$$

Here, the term c_x indicates the true positive, d_y defines the true negative, f_a represents the false negative, and e_z denotes the false positive.

To further enhance the performance of classification, a novel loss function, named MSE MSE and categorical Cross entropy CLF, is introduced an as another innovative element to the proposed AMS-ERNN-NL, which has been formulated through Eq. (12) and Eq. (13). It also paves the way to provide optimal prediction results since it use the balanced error penalization strategy, which combines the power of squared error and cross entropy formulations. It enhances the sensitivity of the model to both large and small deviations, thus representing severe ECG abnormalities and minor deviations that carry early diagnostic significance.

$$MSE = \frac{1}{q} \sum_{l}^{q} \left(b_l - \overline{b_l}\right)^2 \tag{12}$$

Here, the terms q, b_l and $\overline{b_l}$ are the total images, actual and predicted labels.

$$CLF = -\sum_{l=1}^{q} \sum_{m=1}^{n} b_{lm} \log\left(\overline{b}_{lm}\right) \tag{13}$$

Here, the total classes is n, while the actual and predicted label when considering the mth class and l^{th} image are indicated by the terms b_{lm} and $\overline{b}_{lm}$.

This novel loss function aids in enhancing the sensitivity of the system to both the large and minor deviations, where the large deviation indicates the severe ECG abnormalities, while the minor deviation explains the significance of early diagnosis. Thus, the combination of temporal feature modeling, adaptive optimization and multi-scale feature retention, along with the loss function, makes the model robust and resilient towards the complicated and noisy data, thus outscoring the existing system in terms of generalization, stability and accuracy. Consequently, the AMS-ERNN-NL framework provides a comprehensive and clinically reliable solution for heart disease prediction, even under challenging real-world ECG conditions. Figure 5 offers the systematic design of the developed AMS-ERNN-NL-based heart disease prediction model.

6 Results and Discussion

6.1 Simulation Setup

Python was used as the implementation platform. To provide an optimal solution and accurate search, the developed optimization model used 10 population, 3 chromosome length, and 50 iterations. The developed system's performance was validated with traditional approaches, which included VGG-16 [22], XGB [23], LSTM [24] and

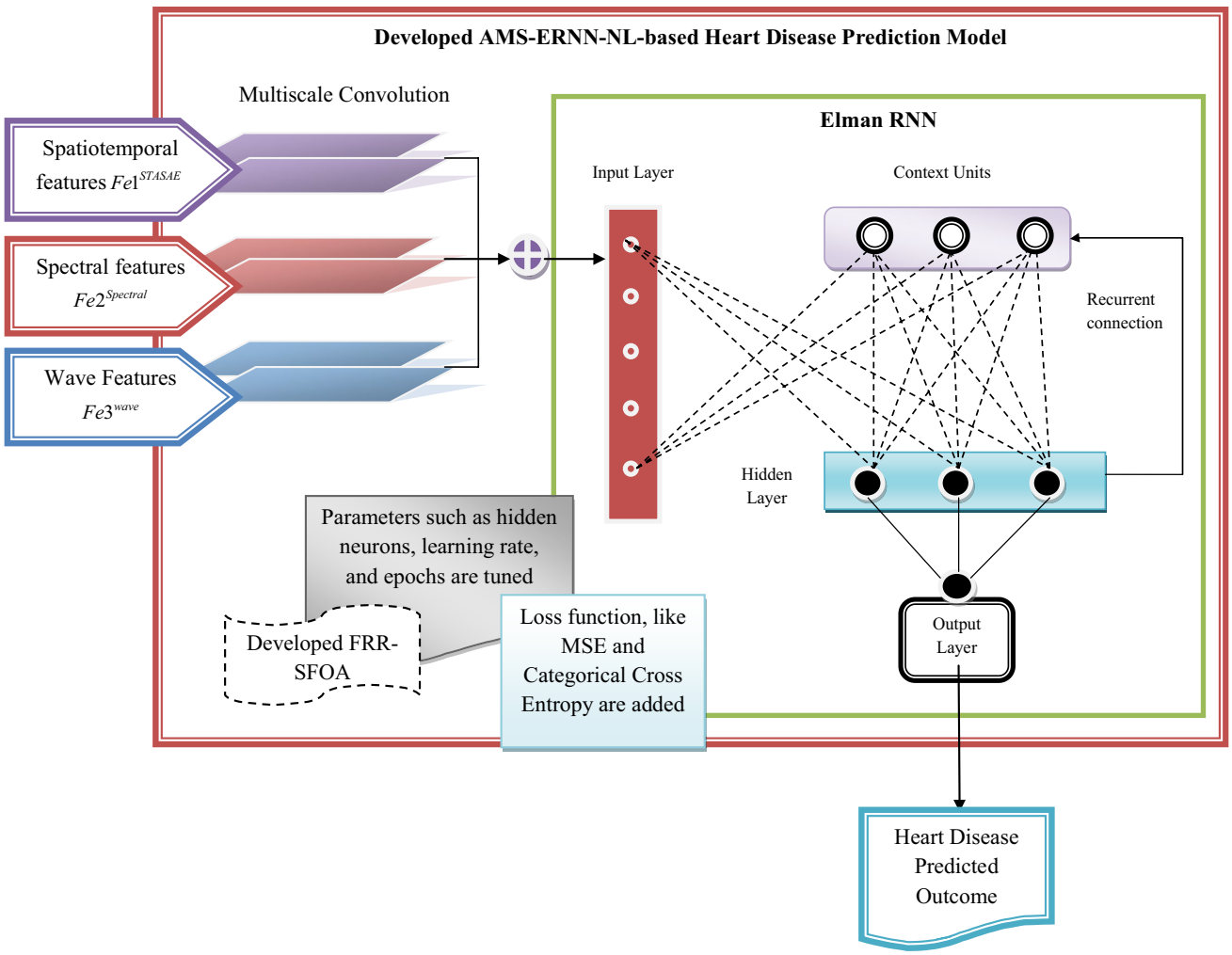


Fig. 5 Systematic Design of the Developed AMS-ERNN-NL-based Heart Disease Prediction Model

ERNN [29], and the optimization algorithm, like Barber Optimization Algorithm (BOA) [30], Orangutan Optimization Algorithm (OOA) [31], Masterpiece Optimization Algorithm (MOA) [32] and SFOA [28], respectively.

6.2 Performance Metrics

The proposed model's performance was validated through the use of the subsequent measures.

1) Accuracy

$$RT_D = \frac{(c_x + d_y)}{(c_x + d_y + e_z + f_a)} \tag{14}$$

2) Negative Predictive Value (NPV)

$$NP_DV = \frac{e_z}{A_F + e_z} \tag{15}$$

3) F1-Score

$$E_{1S} = \frac{2 * c_x}{2 * (c_x + e_z + f_a)} \tag{16}$$

4) False Negative Rate (FNR)

$$NTF = \frac{f_a}{f_a + c_x} \tag{17}$$

5) False Positive Rate (FPR)

$$R^{Pe}F = \frac{e_z}{e_z + d_y} \tag{18}$$

6) Sensitivity

$$se_D = \frac{c_x}{c_x + f_a} \tag{19}$$

7) False Discovery Rate (FDR)

$$QrP = \frac{e_z}{e_z + c_x} \tag{20}$$

8) Balanced Accuracy (BA)

$$QrC = \frac{spec_D + se_D}{2} \tag{21}$$

9) Bookmaker's Informedness (BM)

$$BrI = \sqrt{\frac{spec_D + se_D - 1}{2}} \tag{22}$$

10) Fowlkes-Mallows Index (FM)

$$F_{NM} = \frac{c_x}{c_x + e_z} * \frac{c_x}{c_x + f_a} \tag{23}$$

11) False Omission Rate (FOR)

$$F_O R = \frac{f_a}{f_a + d_y} \tag{24}$$

12) Markedness (MK)

$$MK = PRE + NP_D V - 1 \tag{25}$$

13) Prevalence Threshold (PT)

$$Pt = \frac{se_D + spec_D - 1}{2} \tag{26}$$

Here, the term $spec_D$ is the specificity.

6.3 Convergence Evaluation

The convergence analysis on the proposed system is given in Fig. 6. Here, this examination is carried out with a traditional optimization mechanism, which includes BOA-AMS-ERNN-NL, OOA-AMS-ERNN-NL, MOA-AMS-ERNN-NL, and SFOA-AMS-ERNN-NL. By varying some iteration values, this evaluation is executed to determine the convergence speed of the proposed model. In this evaluation, the existing BOA-AMS-ERNN-NL initially attains the highest cost function, which is 3.50, and then it gradually decreases to 1.65 in its 50th iteration. The OOA shows slightly better stability, dropping from 3.20 to 1.48, while MOA proves a sharper decline from 3.75 to 1.22, representing moderate convergence efficiency. Although the baseline SFOA performs comparatively better and provides convergence from 1.50 to 1.20, it still stabilizes at a higher cost value. But in contrast to the entire existing optimization model, the developed FRR-SFOA-AMS-ERNN-NL attains enhanced optimization through the achievement of a lower cost function, indicating the rapid convergence. The optimal

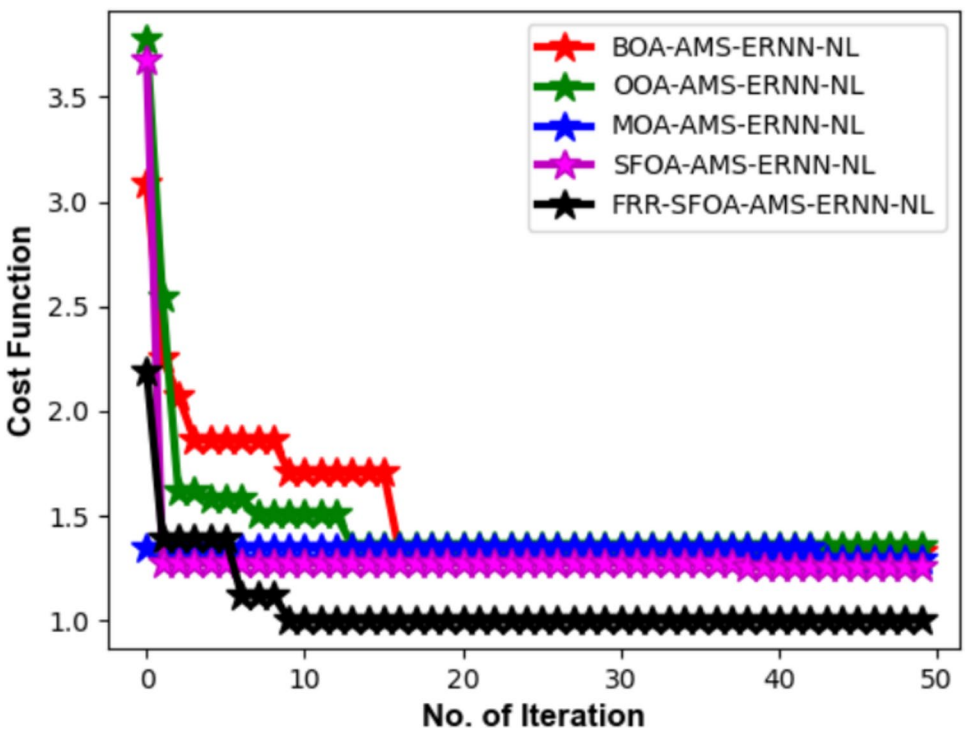


Fig. 6 Convergence Evaluation on the Proposed System

performance of the developed FRR-SFOA-AMS-ERNN-NL starts at 2.30, and within just two iterations, it is reduced to 1.20, and proves a minimal value of 1.00 throughout other iterations. Thus, it was concluded that the developed model has become an optimal choice for enhanced optimization by proving its lower cost function, reduced error score and improved generalization. It further confirms that the fitness-refined randomized strategy applied in the designed FRR-SFOA paves the way to significantly strengthen the optimization process, outperforming traditional and baseline metaheuristics across all iterations.

6.4 Comparison Analysis on the Recommended Approach

Figure 7 visually explains the comparative evaluation of the designed approach with traditional models and optimization mechanisms. This analysis is conducted significantly by altering some of the activation functions, which include Linear, ReLU, Softmax, Sigmoid, and Tanh. This analysis is necessary since it aids in determining how each activation function helps to influence the ECG non-linear patterns present in the system. Some of the activation functions, like ReLU, Sigmoid, and Tanh, are useful to capture the nonlinear relation when compared to Linear and Softmax, where the cardiac abnormalities have different temporal and morphological complexities. When evaluating the varying activation function, it is proven that the developed FRR-SFOA-AMS-ERNN-NL remains active and robust, which makes it suited for delivering high and consistent accuracy, thus demonstrating its well-generalized ability. When analysing the traditional optimization algorithms in terms of linear activation function, it attains enhanced accuracy, ranging from 84.2% for SFOA to 93.2% for the developed FRR-SFOA. For the traditional models in terms of ReLU with F1-score, the scores improve significantly, reaching 85.7% for the proposed model, surpassing LSTM with 79.5% and ERNN with 83.1%. In the Softmax setting, the proposed model again shows superior performance at 81.6%, compared to ERNN with 80.2% and XGB with 76.6%. The Sigmoid activation yields even higher gains, with FRR-SFOA-AMS-ERNN-NL achieving 86.0%, noticeably outperforming ERNN with 82.1% and LSTM with 82.1%. Thus, the results of this enhanced evaluation demonstrate the refined searching behaviour of the designed FRR-SFOA, which leads to attaining superior results by the incorporation of multiscale processing and the novel loss function. This feature also allows the developed system, helping to generalize and maintain stable accuracy effectively regardless of the activation function used.

6.5 ROC Examination of the Developed Mechanism

Figure 8 graphically describes the ROC analysis of the designed model. This evaluation will support offering deeper comparisons to prove the discriminative power of the various approaches widely employed in predicting heart disease. In the figure below, the baseline VGG16 model achieves an AUC of 78.91%, which indicates the limited ability to distinguish the diseased and healthy patterns. XGB proves improved performance by attaining 81.61% AUC, while the LSTM enhances separability through reaching an AUC of 83.75%. The ERNN model shows a stronger learning capacity for temporal ECG variations, achieving an AUC of 86.25%. Moving far away from the entire model, the designed FRR-SFOA-AMS-ERNN-NL outscores the traditional models consistently by attaining the enhanced results in AUC of 88.55%. This reflects the improved specificity-sensitivity, along with the reliability of the designed system. By providing a consistent closer to the left-top corner, the suggested solution proves its strong performance by reducing false positive scores and increasing the true positives. This is possible due to the integration of loss formulation, delivering more accurate heart disease predictions.

6.6 Statistical Analysis on the Recommended Framework

Statistical examination of the proposed system is numerically validated through Table 2. To establish the reliability and stability of the designed FRR-SFOA-AMS-ERNN-NL, this examination is essential, since it offers comprehensive insights into this. When comparing with all the traditional optimization mechanisms, the recommended FRR-SFOA-AMS-ERNN-NL provided optimal results by attaining least best score of 1.0055, indicating

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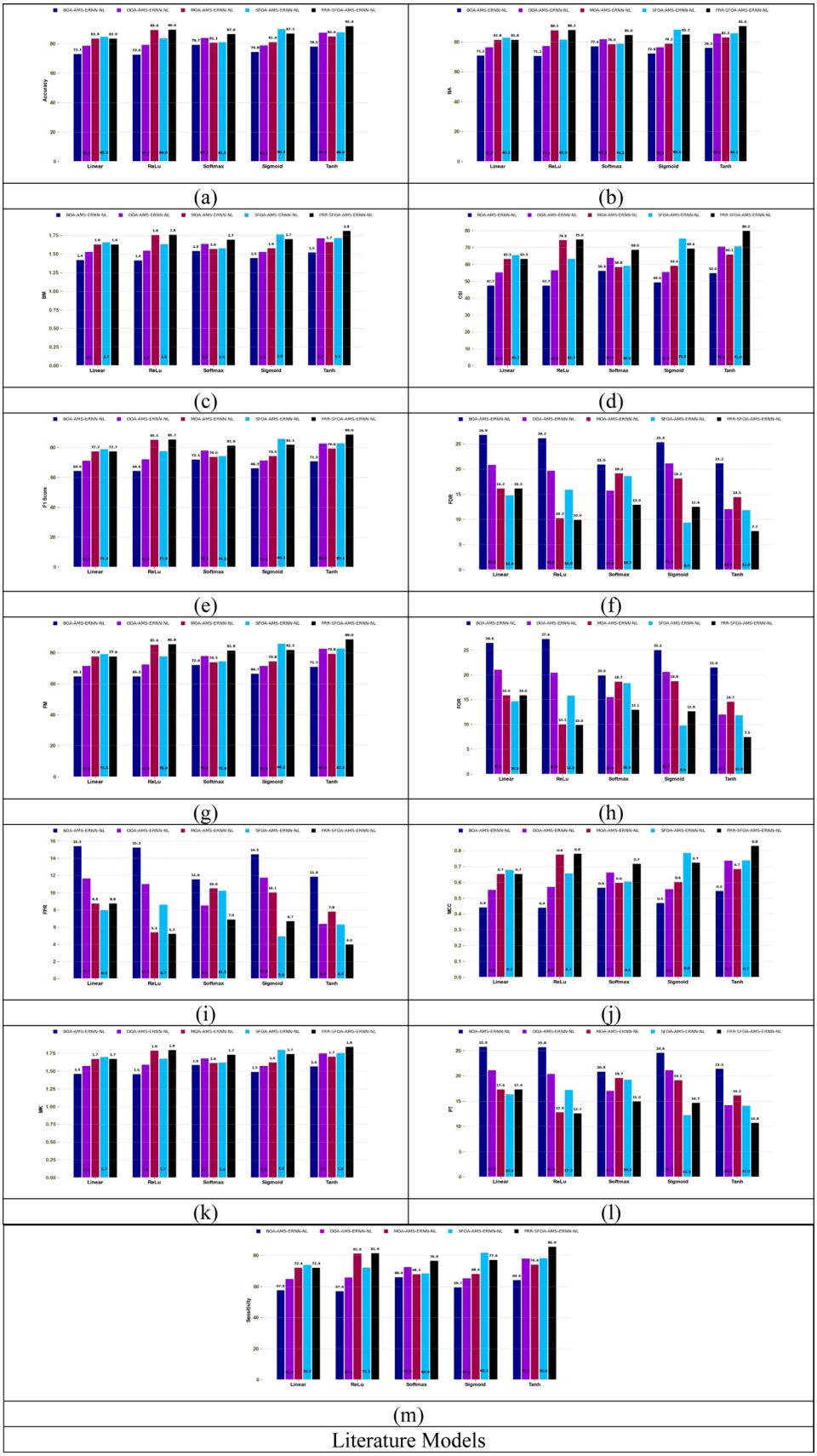


Fig. 7 Comparative Analysis on the Designed System by varying (**a**) Accuracy, (**b**) BA, (**c**) BM, (**d**) CSI, (**e**) F1-Score, (**f**) FDR, (**g**) FM, (**h**) FOR, (**i**) FPR, (**j**) MCC, (**k**) MK, (**l**) PT, and (**m**) Sensitivity

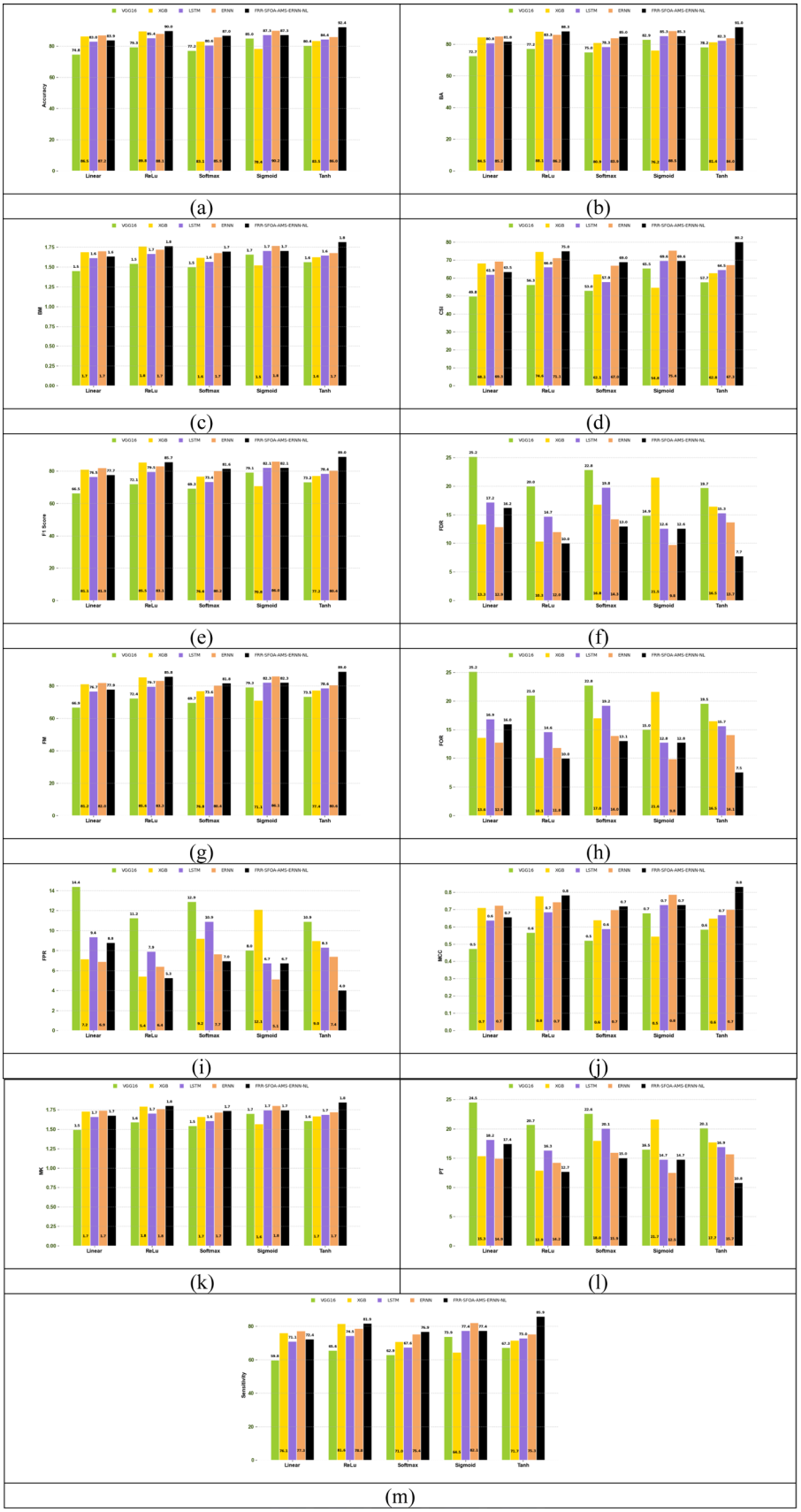


Fig. 7 (continued)

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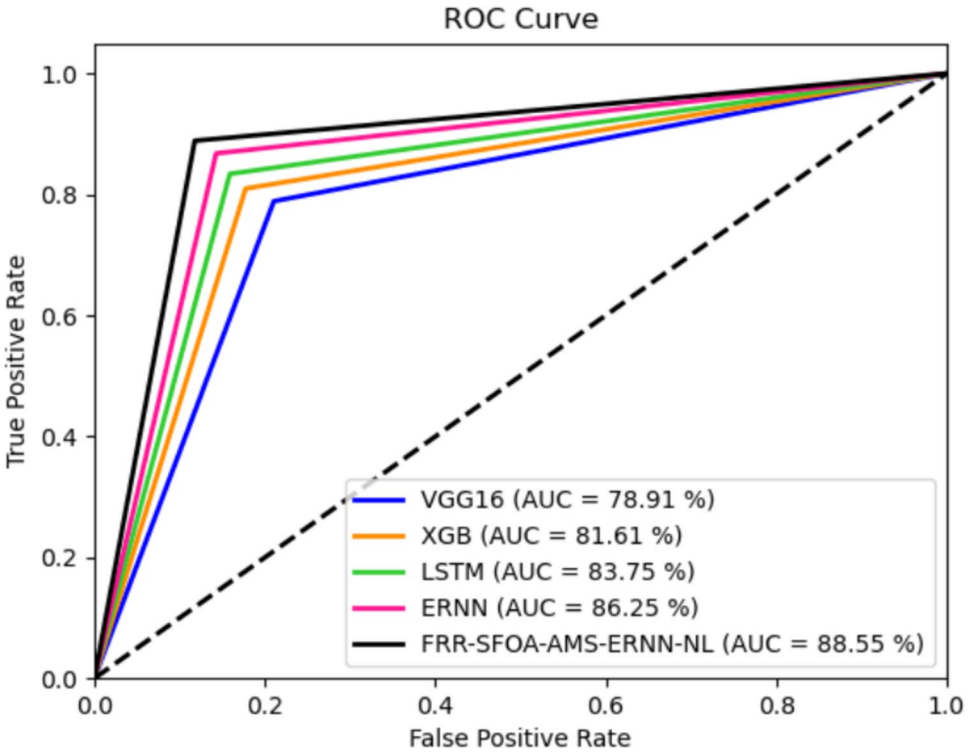


Fig. 8 ROC Evaluation on the Designed Framework

Table 2 Statistical Analysis on the Recommended Framework

TERMS	BOA-AMS-ERNN-NL [30]	OOA-AMS-ERNN-NL [31]	MOA-AMS-ERNN-NL [32]	SFOA-AMS-ERNN-NL [28]	FRR-SFOA-AMS-ERNN-NL
BEST	1.329	1.358	1.292	1.257	1.006
WORST	3.089	3.773	1.348	3.677	2.194
MEAN	1.515	1.472	1.340	1.319	1.075
MEDIAN	1.329	1.358	1.348	1.275	1.006
STD	0.332	0.374	0.020	0.337	0.199

a highly effective cost minimization throughout optimal runs. Its MEAN cost of 1.0754 is which is also the lowest among all algorithms, demonstrating consistent performance. Furthermore, the MEDIAN value aligns with the BEST value of 1.0055, confirming that the proposed optimizer maintains stable convergence across iterations. Although the FRR-SFOA-AMS-ERNN-NL shows a slightly higher WORST value of 2.1939 owing to exploration-driven randomness, its standard deviation of 0.1986 remains significantly lower than BOA-AMS-ERNN-NL with 0.3318, OOA-AMS-ERNN-NL with 0.3741, and SFOA-AMS-ERNN-NL with 0.3370, demonstrating enhanced stability and reduced fluctuation. In comparison, traditional optimizers such as BOA-AMS-ERNN-NL and OOA-AMS-ERNN-NL present higher WORST values, where these are up to 3.77, indicating unreliable convergence. Finally, all the collected and given information shows the evidence to confirm that the FRR-SFOA-AMS-ERNN-NL beat the traditional models more accurately and consistently, thus providing an enhanced converging quality and robustness to tune the heart-disease prediction parameters.

6.7 Numerical Evaluation by Varying Batch Size

Table 3 describes the numerical assessment of the suggested solution. The batch sizes, like 4, 8, 16, 32 and 48, are used for this validation. Here, the measures, including accuracy, specificity, FPR, NPV, etc., are used to showcase

Table 3 Numerical Evaluation by Varying Batch Size

TERMS/Batch Size	BOA-AMS-ERNN-NL [30]	OOA-AMS-ERNN-NL [31]	MOA-AMS-ERNN-NL [32]	SFOA-AMS-ERNN-NL [28]	FRR-SFOA-AMS-ERNN-NL
Accuracy					
4	75.192	85.438	84.698	85.680	92.363
8	84.725	88.315	85.478	84.375	91.515
16	82.654	76.093	82.681	88.127	89.377
32	81.565	81.014	80.866	88.275	88.275
48	88.167	84.174	85.666	86.218	90.440
Specificity					
4	85.741	92.184	91.703	92.288	95.940
8	91.725	93.714	92.152	91.550	95.557
16	90.432	86.351	90.454	93.733	94.359
32	89.971	89.558	89.444	93.766	93.766
48	93.755	91.504	92.194	92.574	95.018
FPR					
4	14.259	7.816	8.297	7.712	4.060
8	8.275	6.286	7.848	8.450	4.443
16	9.568	13.649	9.546	6.267	5.641
32	10.029	10.442	10.556	6.234	6.234
48	6.245	8.496	7.806	7.426	4.982
NPV					
4	75.313	85.397	84.712	85.680	92.457
8	84.732	88.403	85.498	84.349	91.529
16	82.735	76.180	82.755	88.080	89.411
32	81.424	80.960	80.839	88.282	88.282
48	88.120	84.066	85.760	86.244	90.399
MCC					
4	0.481	0.688	0.672	0.693	0.832
8	0.673	0.747	0.688	0.666	0.815
16	0.630	0.499	0.631	0.744	0.769
32	0.610	0.598	0.595	0.746	0.746
48	0.744	0.662	0.692	0.704	0.792
BM					
4	1.460	1.667	1.652	1.672	1.819
8	1.652	1.729	1.668	1.645	1.799
16	1.609	1.478	1.610	1.725	1.752
32	1.588	1.576	1.573	1.728	1.728
48	1.725	1.641	1.672	1.684	1.775
BatchSize	VGG16 [22]	XGB [23]	LSTM [24]	ERNN [29]	FRR-SFOA-AMS-ERNN-NL
Accuracy					
4	79.925	85.774	83.340	87.535	92.363
8	76.980	85.370	86.258	86.944	91.515
16	81.565	79.347	81.296	88.019	89.377
32	87.535	78.540	79.589	85.357	88.275
48	82.506	87.589	90.440	88.719	90.440
Specificity					
4	88.784	92.281	90.859	93.428	95.940
8	86.970	92.102	92.690	92.936	95.557
16	89.794	88.571	89.607	93.647	94.359
32	93.298	87.963	88.617	92.137	93.766
48	90.445	93.415	95.018	94.053	95.018

Table 3 (continued)

BatchSize	VGG16 [22]	XGB [23]	LSTM [24]	ERNN [29]	FRR-SFOA-AMS-ERNN-NL
FPR					
4	11.216	7.719	9.141	6.572	4.060
8	13.030	7.898	7.310	7.064	4.443
16	10.206	11.429	10.393	6.353	5.641
32	6.702	12.037	11.383	7.863	6.234
48	9.555	6.585	4.982	5.947	4.982
NPV					
4	79.992	85.841	83.401	87.455	92.457
8	77.007	85.377	86.184	87.031	91.529
16	81.626	79.246	81.384	87.999	89.411
32	87.596	78.560	79.609	85.317	88.282
48	82.473	87.555	90.399	88.685	90.399
MCC					
4	0.575	0.694	0.644	0.731	0.832
8	0.517	0.686	0.705	0.718	0.815
16	0.608	0.565	0.603	0.741	0.769
32	0.731	0.548	0.569	0.686	0.746
48	0.628	0.732	0.792	0.756	0.792
BM					
4	1.554	1.674	1.623	1.712	1.819
8	1.496	1.666	1.685	1.699	1.799
16	1.587	1.543	1.581	1.722	1.752
32	1.712	1.526	1.547	1.666	1.728
48	1.606	1.713	1.775	1.737	1.775

the strong performance of the recommended approach when considering such significant measures. The FPR-based evaluation offers distinct features to know about the performance of each model, thus helping to avoid the misclassification of healthy individuals as heart disease patients. When it comes to all the values of batch size, the FRR-SFOA-AMS-ERNN-NL consistently achieves the lowest FPR, thus demonstrating enhanced reliability. At batch size 4, it records an FPR of only 4.06%, significantly lower than BOA-AMS-ERNN-NL with 14.26%, OOA-AMS-ERNN-NL with 7.82%, MOA-AMS-ERNN-NL with 8.29%, and SFOA-AMS-ERNN-NL with 7.71%. This trend improves further at batch size 8, where FRR-SFOA-AMS-ERNN-NL reaches an impressively low 4.43%, outperforming MOA-AMS-ERNN-NL with 7.85% and far surpassing BOA-AMS-ERNN-NL with 8.27% and OOA-AMS-ERNN-NL with 6.28%. This consistency of the proposed model in offering minimized false positive values indicates its ability to reduce unwanted clinical alarms, enhancing diagnostic confidence and avoiding misclassification scores more effectively than all competing models.

7 Conclusion

This work developed an enhanced model, which helped to bridge the gap present in between the collected ECG signals and health insights. Through the use of STA-SAE, the developed model extracted the essential spatio-temporal features from the collected ECG signals. Moreover, wave and spectral features were also extracted for the effective and accurate prediction process. Then, AMS-ERNN-NL was applied, which helped to classify the heart disease by capturing both long-term patterns and instantaneous morphological data related to the heart activity. The inclusion of a novel loss function and the enhanced optimization mechanism, named FRR-SFOA, ensured that the proposed model attained high accuracy without compromising the model's robustness towards overfitting. Finally, the designed system predicted heart disease. When evaluating the 4th batch size, the accuracy

of the FRR-SFOA-AMS-ERNN-NL was 92.36%, where the traditional optimization models, such as BOA-AMS-ERNN-NL, OOA-AMS-ERNN-NL, MOA-AMS-ERNN-NL, and SFOA-AMS-ERNN-NL, provided the values were 75.19%, 85.43%, 84.69% and 85.68%. Thus, the results proved that the developed model provided a dynamic tool to offer early detection and monitoring, which assisted in identifying the evolving cardiovascular conditions, thus minimizing the diagnostic errors and maximizing the patient outcomes in clinical settings.

Though the developed system holds technical innovations, the complexity of this is greater since the model processes the signals using the multi-scale element. Then, the performance of the model is more sensitive to the noisy data and artifacts, which can degrade the performance significantly. Thus, the future work will focus on using Explainable AI (XAI) to help provide visual evidence of the specific waveform patterns. Finally, implementing federated learning protocols could allow for multi-institutional model training without compromising patient data privacy.

Author Contribution Vaanathi Siva, Palanisamy C, designed the model, computational framework and carried out the implementation. Vaanathi Siva performed the calculations and wrote the manuscript with all the inputs. Vaanathi Siva, Palanisamy C, discussed the results and contributed to the final manuscript.

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Data Availability No datasets were generated or analysed during the current study. In case of benchmark data: The data underlying this article are available in ECG Heartbeat Categorization Dataset, at https://www.kaggle.com/datasets/shayanfazeli/heartbeat.

Declarations

Conflict of Interests The authors declare no competing interests.

Ethics Approval Not applicable.

Consent to Publish Not applicable.

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